Surgical Tool Detection with Semi-Supervised Learning

Computer Vision — Surgical Applications, HW1

Authors: Raz Ben-Aharon and Shalev Manassen

1. Exploratory Data Analysis

Our labeled set contained only **61 training** and **10 validation** images (<100 total), all in native 4K. Each frame comes from a surgery video, and our objective was to correctly identify bounding boxes encompassing a gloved hand along with what it holds: **Empty**, **Tweezers**, or **Needle_driver**.

1.1 Visualization of Some Images

Fig. 1 shows a random sample of our labeled training frames with ground-truth boxes (green = Empty, blue = Tweezers, red = Needle_driver).

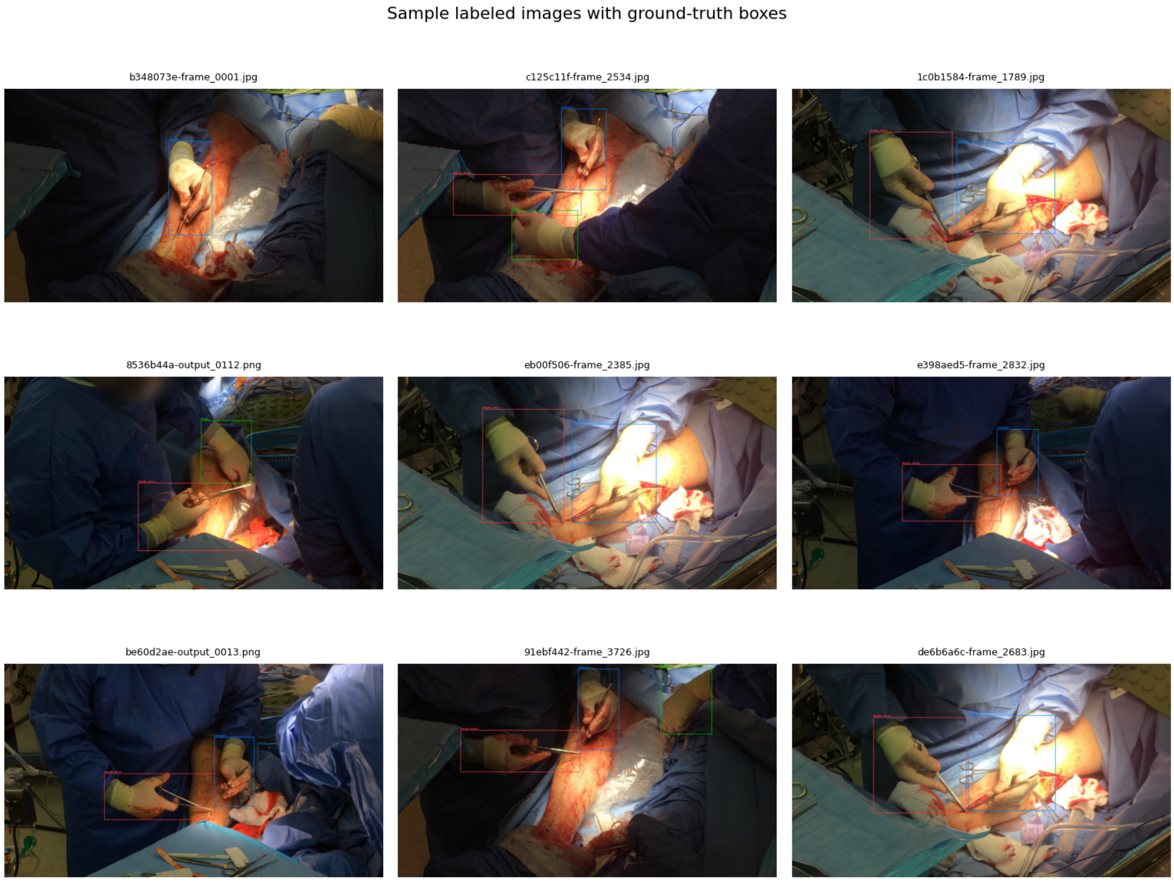

Fig 1. Nine labeled frames with ground-truth boxes overlaid.

1.2 Insights from our Initial Analysis

Before training, we manually inspected the frames and videos. We quickly noticed several challenges: the scene is visually difficult due to specular highlights, blood, and surgical drapes that blend in with the gloves. Furthermore, almost every frame contains exactly one or two boxes (Fig. 2, right). Crucially, the OOD video was filmed with completely different lighting and camera angles, meaning our model would need strong generalization capabilities, not just high ID accuracy.

1.3 Data Distribution

Split	Images	Boxes	Empty	Tweezers	Needle_driver	Boxes/img
train	61	135	26	54	55	2.21
val	10	22	2	10	10	2.20

Empty is the clear minority class in our labeled set (only ≈19% of train boxes), while Tweezers and Needle_driver are roughly balanced. This class imbalance strongly motivated our pseudo-labeling strategy, as frames with empty hands are common in the raw video but scarce in our labeled sample.

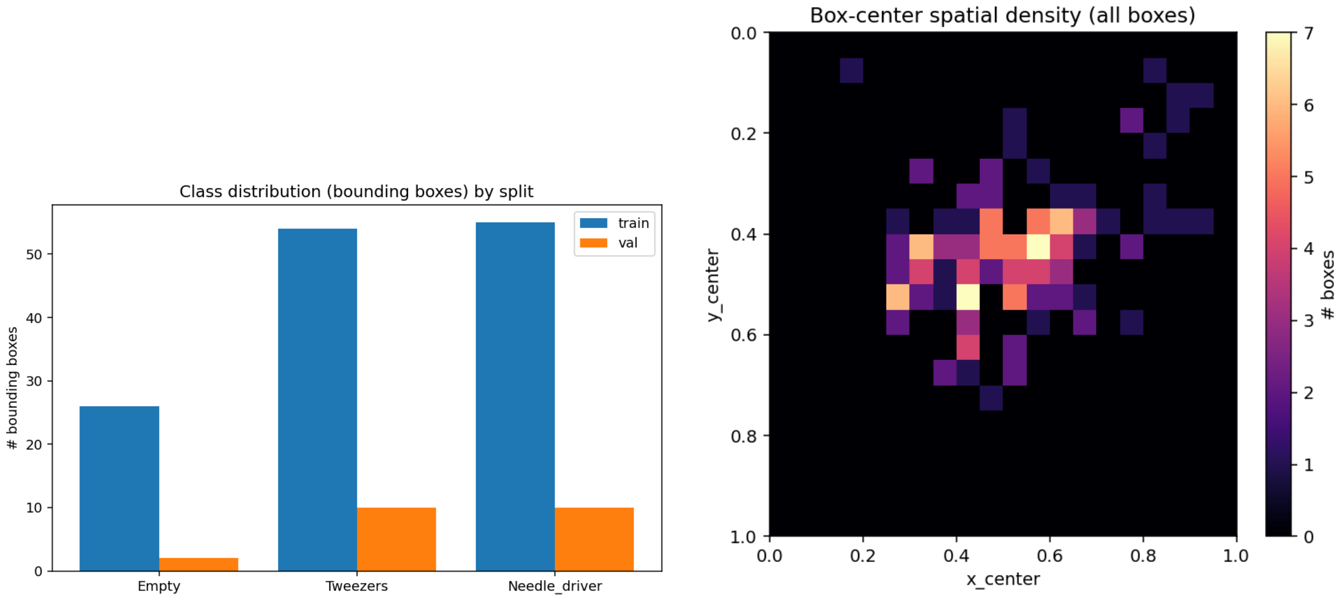

Fig 2. Left: Bounding-box counts showing Empty as the minority class. Right: Spatial density of box centers, matching the top-down camera setup.

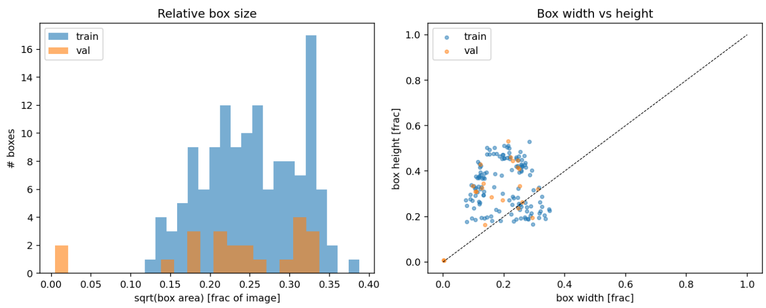

Fig 3. Left: Distribution of relative box sizes. Right: Box width vs. height.

2. Experiments

Our goal was to maximize out-of-distribution (OOD) generalization under a very limited budget of 61 labeled training images. We treated the task as a semi-supervised learning challenge. Because every simulation (or in our case, training run) can be misleading, we decided not to rely solely on in-distribution metrics. We evaluated every model choice by its actual qualitative behavior on the OOD video. Furthermore, we treated **confirmation bias** as a significant risk — naive pseudo-labeling can turn a minor mistake into a systemic failure. Therefore, simply having more pseudo-labeled data is a misleading metric; what matters most is the *quality* of the labels.

2.1 Data Loading & Validation Audit

To start, we had to establish a strong supervised Baseline and audit our data. We manually inspected our validation images and found two erroneous bounding box annotations in ff8c22da-output_0182.png. Because these artifacts represented ≈9% of our tiny validation set, we computed 'cleaned' metrics (val_clean) alongside the official metrics (val_original) to ensure our hyperparameter choices were based on reality.

2.2 Hyperparameter Tuning

We ran a comprehensive hyperparameter sweep, testing different resolutions (640/960/1280), learning rates (0.001/0.003/0.01), and model sizes (YOLO11s/YOLO11m) to find the most robust baseline. We found that higher learning rates (0.003+) caused training collapse, and larger models didn't help. Interestingly, our **Supervised 960** model achieved the highest cleaned validation mAP (0.854), but we ultimately rejected it because manual OOD inspection revealed recurring false positives on surgical gauze.

2.3 Semi-Supervised Learning & Regularization

After establishing our baseline, we began building our semi-supervised models. Our original SSL pipeline sampled frames and kept those where the model was highly confident (≥0.7). However, we quickly saw that systematic teacher errors (such as misclassifying blood-stained gauze as Needle_driver) were being reinforced. To avoid wasting our model's capacity on bad labels, we experimented with **sanity constraints** and **temporal filtering**, requiring predictions to persist across neighboring frames. While this prevented false positives, it severely hurt our recall. To regularize, we relied on weight decay (5e-4), early stopping, and moderate data augmentation (mosaic, color jitter).

2.4 Train + Valid Loss Graph

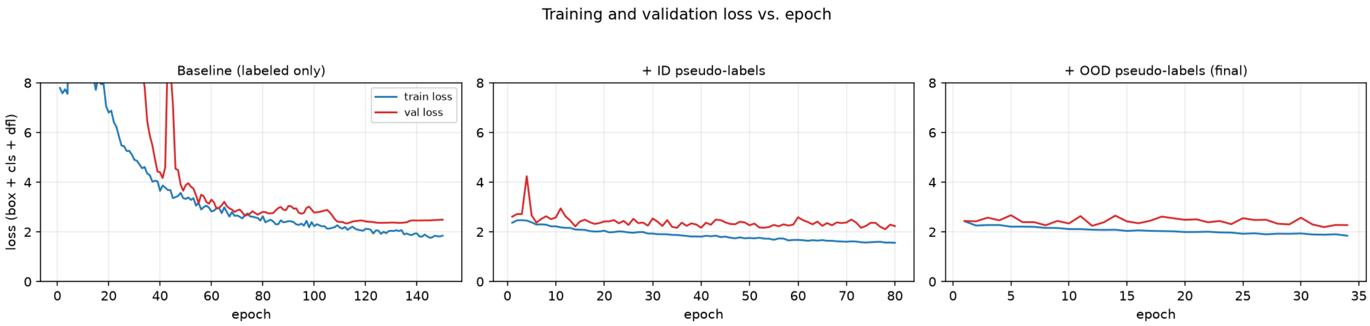

Fig 4. Training (blue) and validation (red) loss vs. epoch. Our Baseline shows an early instability period during LR warmup, then recovers smoothly without overfitting.

2.5 Train + Valid mAP Graphs

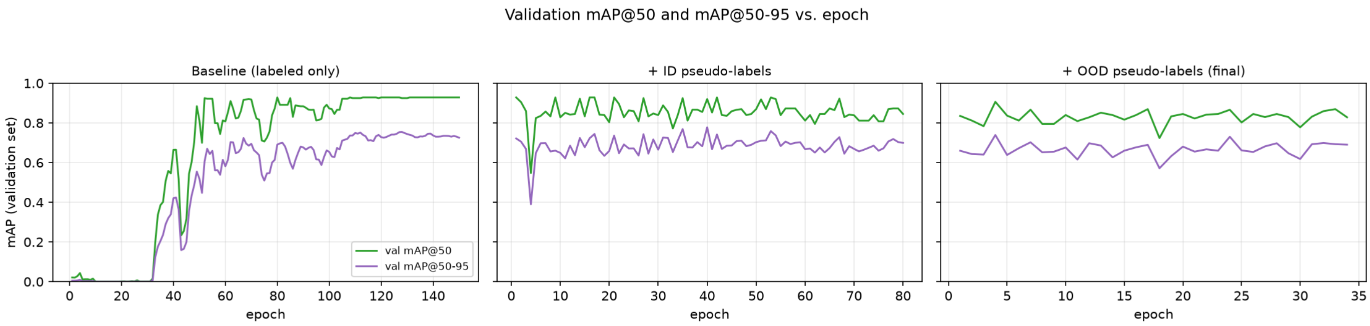

Fig 5. Validation mAP@50 (green) and mAP@50-95 (purple) vs. epoch.

Model	Epoch	Imgs	mAP50-95 orig	mAP50-95 clean	OOD behavior
Baseline (final)	128	61	0.754	0.809	Best balance
ssl_id	40	763	0.782	0.837	87% >2 det
ssl_ood	4	1430	0.740	0.792	90% >2 det
Sup. 960	103	61	0.797	0.854	Gauze FP
Sup. 1280	76	61	0.751	0.805	Flicker
ID P4	11	130	0.749	0.800	81% >2 det
ID temporal	18	627	0.767	0.820	16% empty

ID+OOD comb.	24	750	0.721	0.775	21% empty

Table 1. Key candidates with official and cleaned validation mAP. Notice that our chosen Baseline doesn't have the highest mAP, but rather the best OOD behavior.

3. Discussion and Conclusions

The Danger of Confirmation Bias

Our initial OOD self-training pipeline degraded rather than improved OOD generalization. We discovered that 71.7% of the OOD pseudo-label boxes were classified as Needle_driver, revealing a massive class skew. Manual inspection showed recurring false-positive Needle_driver detections on blood-stained gauze. This is a classic case of confirmation bias: our teacher model made mistakes on the OOD video, and by blindly training on those mistakes, the model became even more confident in its incorrect patterns. We learned that **confidence alone is insufficient** (our P4 policy retained predictions at 0.938 confidence but still failed dramatically).

Finalist Comparison

To ensure our comparisons were reliable, we evaluated our top candidates using automated OOD video diagnostics, tracking metrics like frames with >2 detections and single-frame flickers. We then manually evaluated the finalists on the full 180-second OOD video.

Finalist	det/fr	>2 %	empty %	1-frame tracks	class switches
Baseline / 0.60	1.70	6.6	5.9	145	69
ID temporal / 0.40	1.38	0.5	16.0	92	18
Sup. 1280 / 0.50	1.49	5.1	10.5	288	82

Table 2. Diagnostics on the full 180s OOD video for our 3 finalists.

Our **ID temporal** model was extremely stable (few false positives) but failed to detect real visible hands too often. Our **Supervised 1280** model had reasonable coverage but suffered from excessive temporal flicker. The **Baseline** model offered the best overall balance of precision and recall on the unseen domain.

Conclusion

Ultimately, we selected our supervised Baseline (at confidence 0.60) as the final detector because it produced the best overall OOD video behavior, despite not achieving the highest validation mAP. Naive pseudo-label self-training introduced confirmation bias and severe false positives. Stricter confidence thresholds alone were insufficient, while sanity and temporal filtering improved prediction stability but reduced recall. Our experiments demonstrate that higher validation metrics and larger pseudo-labeled datasets do not necessarily translate into better real-world generalization. Sometimes, a well-regularized baseline is the most robust choice.